

2	(a)	Gravitational field strength is defined as the <u>force per unit mass</u> acting on a small mass placed at a point in the gravitational field.	A1
	(b)	For the same difference in gravitation potential, <u>the distance between the potential lines is increasing</u> as the distance from the surface of Mars, show that gravitation field strength is decreasing.	A1
	(c)	Loss in gravitational potential energy = gain in kinetic energy of the satellite $(90)(-6.0 - (-8.0)) \times 10^6 = (1/2)(90)v^2$ $v = 2000 \text{ ms}^{-1}$	C1 A1
	(d)(i)	Centripetal force of the satellite is provided by the gravitation force. $mr\left(\frac{2\pi}{T}\right)^2 = \frac{GMm}{r^2}$ $T = \sqrt{\frac{4\pi^2 r^3}{GM}}$ $= \sqrt{\frac{4\pi^2 (3.4 \times 10^6 + 1.7 \times 10^7)^3}{G(6.4 \times 10^{23})}}$ $= 88607 \text{ s} = 24.6 \text{ h}$	C1 C1 (correct radius) A1
	(d)(ii)	$\text{KE} = \frac{1}{2}mr^2\omega^2$ $= \frac{1}{2}(90)(3.4 \times 10^6 + 1.7 \times 10^7)^2 \left(\frac{2\pi}{88607}\right)^2$ $= 9.42 \times 10^7 \text{ J}$ Alternative $\text{KE} = \frac{GMm}{2r}$ $= \frac{G(6.4 \times 10^{23})(90)}{2(3.4 \times 10^6 + 1.7 \times 10^7)}$ $= 9.42 \times 10^7 \text{ J}$	C1 A1
	(d)(iii)	Due to work done against friction, <u>total energy of the satellite decreases, causing the satellite to fall towards Mars (or radius of orbit decreases).</u> <u>As the satellite fall towards Mars</u> , it experiences a loss in gravitational potential energy and an <u>increase in kinetic energy</u> . (Or Since $\text{KE} = \text{GMm}/2r$ as radius decreases, KE increases)	B1 B1

3	(a)	Despite both being at 100 °C, steam transfers more (heat or thermal) energy	B1
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